

Aarya Arban

T11 - 05

Assignment No. 9

Aim: Add a service worker to implement Push notifications in web app

Push Notification – Write-up

Introduction

Push notifications are messages sent by servers to client applications (like mobile apps or websites), even when the application is not currently in use. They are used to inform, engage, and re-engage users with timely updates, promotions, or alerts.

What are Push Notifications?

Push notifications are a form of **real-time communication** between the server and the user. Unlike traditional messages that require the user to open the app to receive updates, push notifications are **delivered instantly** to the user's device, even when the app is closed.

How Push Notifications Work

1. User Subscription

- The user grants permission to receive push notifications.
- A unique **subscription token** or **device token** is generated for the user.

2. Server-Side Trigger

- The server (or backend) sends a push message using the token.
- A push service like **Firebase Cloud Messaging (FCM)** or **OneSignal** is used.

3. Service Worker (For Web)

- On web applications, a **Service Worker** runs in the background and listens for incoming push events.
- It displays the notification using the **Notification API**.

4. Notification Display

- A system notification is shown to the user.
- Tapping the notification can open a specific part of the app or website.

Components Involved

- **Push Service (e.g., Firebase Cloud Messaging)**
- **Service Worker (Web only)**
- **Notification API**
- **User Device or Browser**

Sample Code (Web Push Notification)

```
self.addEventListener("push", function(event) {  const
data = event.data.json();
  self.registration.showNotification(data.title, {
body: data.body,
  icon: "icon.png"
});
});
```

Use Cases

- Social media notifications (likes, comments)
 - News and weather alerts
 - Reminders for to-do or calendar apps
 - Promotional offers and discounts
 - Order status or delivery updates
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Benefits

- Keeps users informed in real-time
- Improves user engagement and retention
- Works even when the app is not active
- Allows personalized, targeted communication

Challenges

- Users may disable notifications if they receive too many
 - Requires careful handling of permissions
 - Platform-specific implementation (Android, iOS, Web)
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Conclusion

Push notifications are a powerful communication tool that bridges the gap between applications and users. When implemented thoughtfully, they can significantly enhance user experience and interaction.

LO Mapping: LO6